

ACTIVITY 22 — FACILITATOR GUIDE**SPC Control Chart and Statistical Process Analysis**

TYPE Workshop	DURATION 60 minutes	MAPS TO LO5	TOPIC 5 - Monitor and Control the Project
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Purpose of this activity

By the end of this activity learners will be able to plot a real process over time and compute control limits from the data itself.

What learners produce

control chart, stability verdict, capability summary — submitted to artifacts/.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- Laptop with internet access for the live problem-solving tools
- The course slides for the framework and worked example

Set-up

Groups of 3 to 4. Distribute one worksheet per learner and one checklist per group; each group needs one laptop for the calculations.

Scenario / briefing material

Contoso Learning Pte Ltd is a training provider with 14,000 active learners. Its course registration portal is 9 years old. Registration takes a median of 11 minutes and 34% of attempts are abandoned. Learner communications consume 24 staff-hours per week across 3 administrators. 62% of learner traffic is mobile, against a portal that is not responsive. Legacy consent screens do not meet current PDPA requirements. The project modernises course registration and learner communications.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Take the measurement data. Contoso measures median registration completion time every working day. Charter success criterion SC-1 requires the completed system to achieve under 4.0 minutes, down from the 11-minute legacy baseline.	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates into every later step.
2	Compute the centre line and control limits. This is an individuals chart (X-mR), used when each measurement is a single value rather than a subgroup average. The control limits come from the moving range, not from the standard deviation.	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.
3	Plot the chart. Enter the full 20-point series in order:	Ask what the number would have to be before the decision flips. That is the exam's favourite

		question.
4	Apply the out-of-control rules. A single point outside the limits is only one of several signals. Apply the full set - these are the Western Electric / Nelson rules, and the exam expects the rule of seven in particular.	Name the assumption behind the entry. Unstated assumptions are what the exam punishes.
5	Separate the causes and decide the response	Pick one entry and follow it back to its source document. Broken traces surface immediately.
6	Assess capability against the charter target. Now bring in the specification limit, which has been absent so far and deliberately so.	Ask what the number would have to be before the decision flips. That is the exam's favourite question.
7	Write the quality verdict and actions. Create artifacts/22-stability-verdict.md:	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates into every later step.
8	Answer the exam-style scenarios	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.

Suggested timing

Minutes	Phase
6 min	Brief the scenario, confirm the inputs and assign roles
22 min	Work the first half of the steps individually, then pool as a group
23 min	Complete the remaining steps and finalise the deliverable
9 min	Group share-back, facilitator debrief and exam-style linkage

Total: 60 minutes

Discussion & decision prompts

1. Plot a real process over time and compute control limits from the data itself — how did your group decide, and what would change that decision?
2. Apply the out-of-control rules, including the rule of seven, to detect signals — how did your group decide, and what would change that decision?
3. Separate common cause from assignable (special) cause variation, and choose the correct response to each — how did your group decide, and what would change that decision?
4. Your centre line is 4.053, MR-bar is 0.3286, sigma is 0.2913, and your limits are 3.179 and 4.927. If your limits are much wider, you computed them across all 20 points and have hidden the signal — can every member of your group defend it?
5. Where would spc control chart and statistical process analysis have saved you time or money on a real project you have worked on?

What good looks like

- Your centre line is 4.053, MR-bar is 0.3286, sigma is 0.2913, and your limits are 3.179 and 4.927. If your limits are much wider, you computed them across all 20 points and have hidden the signal.
- You used MR-bar / 1.128 to estimate sigma, not the sample standard deviation. Using the sample SD of the stable period (0.2066) would give limits of 3.43 to 4.68 - narrower, and not the standard individuals-chart method.
- You checked the rule of seven explicitly on days 1-15 and found the longest run to be 2, concluding no violation. Simply asserting "no rule violations" without the run analysis is not the check.
- You identified days 16-20 as assignable cause AND named a specific, dated, plausible cause with a magnitude that matches the observed shift.

- You stated that days 1-15 were in control and still failed the specification. If your verdict is "the process was fine until day 16", you have missed the central point of the lab.
- Your Cpk is negative, and you concluded that the process must be re-centred rather than merely tightened.
- Your action list addresses both the assignable cause and the underlying capability problem, and your post-action forecast is stated honestly rather than claimed as capable.
- You can explain why demanding an explanation for a single below-average day in a stable process makes performance worse.